

PARALLEL SESSION #5:

DIGITAL TRANSFORMATION & SUSTAINABILITY



DATE

October 24th, 2025



TIME

09.00 - 11:50 am



LOCATION

ROOM 311 LECTURE HALL

VGU CAMPUS

THOI HOA WARD,
HO CHI MINH CITY



Detailed Agenda:

<https://conference.vgu.edu.vn/agenda>



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Parallel Session #5: Digital Transformation & Sustainability

AGENDA

 FRIDAY
 OCT.24.2025
 09:00 AM - 12:00 PM

9:00 AM	REGISTRATION	Welcome & Registration
9:30 AM	MAIN CONFERENCE	Digital Transformation at Universities <i>Buu-Kieu Lam</i>
09:50 AM	MAIN CONFERENCE	AI for Green energy: Reinforcement Learning and Its Application <i>Tuan-Ngoc Nguyen & Dinh-Huy Truong</i>
10:10 AM	MAIN CONFERENCE	Sustainable Software: What is it and why should we care? <i>Manuel Clavel</i>
10:30 AM	TEA BREAK	
10:50 AM	MAIN CONFERENCE	Digital Solutions for Digital and Green Economy: Case Studies <i>Phuc-Hong Tran</i>
11:10 AM	MAIN CONFERENCE	Digitalizing ancient Vietnamese script: Nom character recognition from limited annotation datasets <i>Tuan-Cuong Nguyen</i>
11:30 AM	MAIN CONFERENCE	Digital Twins: from Twinning Physical Assets to Twinning Business Processes. <i>Lam-Son Le</i>
11:50 AM	LUNCH	



CONFERENCE: FOSTERING GREEN SCIENCE AND INNOVATION
PARALLEL SESSION #5: DIGITAL TRANSFORMATION & SUSTAINABILITY
TIME: 09:00 AM - 11:50 AM DATE: FRIDAY, OCT.24.2025
ROOM 311 LECTURE HALL, VGU CAMPUS, THOI HOA, HO CHI MINH CITY

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from Twinning Physical Assets to Twinning Business Processes.

Lam-Son Le

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Parallel Session #5: Digital Transformation & Sustainability

DIGITAL TRANSFORMATION AT UNIVERSITIES

ABSTRACT

The presentation provides the requirements for digital transformation from both international accreditation bodies and within Vietnam. It serves the university's digital pillars, guiding action plans and its goals for 2030. The talk explains the stages of the digital journey—digitization, digitalization, and digital transformation—and emphasizes the importance of data and data maturity. It also highlights the key components for a sustainable digital transformation, emphasizing that digital competence and leadership are crucial. The presentation concludes with a discussion on common challenges and lessons learned for the university towards success.

**Ms. Bửu-Kiều Lâm**Deputy Head of
Information Technology Department**Speaker**

BIO

Ms. Kieu holds a Master of Science (M.Sc.) in Computer Science from the Technische Universität Darmstadt in Germany. With a career spanning over a decade in IT and project management, Ms. Kieu specializes in the intersection of Digitalization, Enterprise Resource Planning (ERP), and Process Improvement. Her skills are critical to creating efficient, modern operating environments, which are foundational for sustainable and green initiatives.

Ms. Kieu is the Deputy Head of the IT Department at Vietnamese-German University (VGU), where she leverages her extensive experience to drive digital maturity and strategic planning across the institution. Her passion lies in harnessing digital tools and strategic planning to transform processes, demonstrating a clear commitment to fostering a forward-thinking and sustainable future through technology.

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Parallel Session #5: Digital Transformation & Sustainability

AI FOR GREEN ENERGY: REINFORCEMENT LEARNING AND ITS APPLICATION

ABSTRACT

Reinforcement Learning (RL) has emerged as a transformative approach for addressing complex sustainability challenges across energy, mobility, manufacturing, and resource management. This study provides a comprehensive overview of RL fundamentals, key algorithms, and their applications in advancing digital transformation with a green energy lens. Case studies highlight RL's effectiveness in smart energy systems—optimizing microgrids, battery energy storage, and grid topology control—leading to improved resilience, efficiency, and reduced emissions. Applications in building automation and demand response demonstrate significant energy and cost savings, while RL-driven traffic management and green mobility reduce congestion and enhance urban sustainability. We also discuss advanced paradigms such as multi-objective RL, graph-based RL, and integration with digital twins and IoT for scalable, real-time decision-making. Finally, the study addresses open challenges including scalability, data availability, interpretability, and the environmental footprint of RL itself. Overall, RL represents a powerful decision-making engine for enabling adaptive, efficient, and sustainable operations, aligning with global net-zero and smart city goals.



Dr. Đình-Huy Trương

Lecturer in Computer Networks &
Internet of Things

and

Mr. Tuấn-Ngọc Nguyễn

Lecturer in Computer Networks &
Internet of Things

Speaker

BIO

Nguyễn Tuấn Ngọc is a graduate student in the Information Technology and Science program at the Vietnamese-German University (VGU). His research interests center on Reinforcement Learning and its applications in sustainable energy optimization. He possesses practical experience in software engineering, data science, and cybersecurity.

Dr. Huy is a Lecturer in Computer Networks & Internet of Things in the Computer Science program at the Vietnamese-German University (VGU). His research focuses on energy optimization for IoT networks through the application of Artificial Intelligence and Deep Learning techniques.



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Parallel Session #5: Digital Transformation & Sustainability

SUSTAINABLE SOFTWARE: WHAT IS IT AND WHY SHOULD WE CARE?

ABSTRACT

In this talk, I will first discuss the meaning of “sustainability” in the case of software development and review the reasons why “sustainable software” matters.

Next, I will analyze the usage of Generative AI tools for software development and their impact on the environment.

Finally, I will discuss the benefits of using Cloud Computing for sustainable software development. Throughout the talk, I will highlight some “best practices” for sustainable software development to be included in a Software Engineering Curriculum.



Assoc. Prof. Garcia Clavel Manuel

Senior Lecturer in Programming
Languages and Methods

Speaker

BIO

Manuel Clavel received his bachelor's degree in Philosophy from the Universidad de Navarra (Spain) in 1992, and his Ph.D from the same university in 1998. Currently, he is Senior Lecturer in Programming Languages and Methods at the Faculty of Engineering of Vietnamese-German University (Binh Duong, Vietnam). Previously, he was Associate Research Professor at the IMDEA Software Institute in Madrid (Spain), as well as Associate Professor at the Universidad Complutense de Madrid (Spain) and Visiting Professor at ETH Zurich (Switzerland). During his doctoral studies, he was International Fellow at the Computer Science Laboratory of SRI International (1994-1997) in California (USA) and a Visiting Scholar at the Computer Science Department of Stanford University (1995-1997). His research focuses on rigorous, tool-supported model-driven software development, including: modeling languages, model-driven security, model-driven privacy, model-driven code-generation, and model-driven automated-test generation. Related interests include specification languages, automated deduction, and theorem proving.

✓ **10:10 – 10:30AM, FRIDAY, OCT 24TH 2025**

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Parallel Session #5: Digital Transformation & Sustainability

DIGITAL SOLUTIONS FOR DIGITAL AND GREEN ECONOMY: CASE STUDIES

ABSTRACT

This presentation showcases case studies from TMA Technology Group on how digital solutions contribute to Digital and Green Economy development in Vietnam and other countries, across industries such as agriculture, public health, and manufacturing. It also highlights collaboration models between enterprises and universities in R&D and product commercialization, along with best practices for applying digital technologies to support community, social, and economic development

**Mr. Phúc-Hồng Trần**

Senior Vice President, TMA Technology Group
Vice President, Vietnam Digital Technology
Alliance (VNITO)

Speaker

BIO

Mr. Hong Tran brings over 27 years of expertise in innovation, digital transformation, R&D, international business, and technology consulting, serving global companies across Asia, Australia, Europe, and North America. He is currently Senior Vice President at TMA Technology Group—a leading Vietnamese software company with nearly 4,000 engineers and clients across 30 countries. In addition, he serves as Managing Director of TMA Innovation and Vice President of the Vietnam Digital Technology Alliance (VNITO).

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Parallel Session #5: Digital Transformation & Sustainability

DIGITALIZING ANCIENT VIETNAMESE SCRIPT: NOM CHARACTER RECOGNITION FROM LIMITED ANNOTATION DATASETS

ABSTRACT

This research proposes a comprehensive solution for the recognition and digitization of Nom characters, an ancient Vietnamese writing system. The study leverages Ideographic Description Sequences (IDS) to address zero-shot and few-shot learning challenges in Nom character recognition. Our approach also apply the IDS representation for effective clustering and similarity-based search. The system includes automatic segmentation of Nom characters from documents using object detection and image segmentation models. Additionally, we develop a user-friendly web application for researchers to collect, preprocess, label, recognize, and query Nom character data. By integrating these components, our system aims to significantly advance the preservation and accessibility of Vietnam's valuable historical documents, while overcoming the limitations of current Nom digitization efforts that struggle with the vast diversity of characters and limited labeled datasets.



Dr. Tuấn-Cường Nguyễn

Lecturer in Information Systems and
Data Science

Speaker

BIO

Dr. Nguyen Tuan Cuong is a lecturer in Information Systems and Data Sciences at Vietnamese-German University (VGU). He received his Ph.D. from Tokyo University of Agriculture and Technology (TUAT) in 2017. He had been an assistant professor at TUAT (2017-2021), an associate professor at the WACOM-TUAT joint research laboratory (2022), and a lecturer/researcher at FPT University (2023). He has been working on online and offline handwritten math/Japanese text/English text recognition, clustering and automatic scoring, co-organized VOHTR2018 competition on Vietnamese online handwriting recognition and CROHME2023 competition in recognizing handwritten mathematical expression.



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Parallel Session #5: Digital Transformation & Sustainability

DIGITAL TWINS: FROM TWINNING PHYSICAL ASSETS TO TWINNING BUSINESS PROCESSES

ABSTRACT

Ever since the dawn of digital twin technologies, we have been introduced to advanced analytics and decision support on manufacturing, supply chain management, construction, etc. When it comes to twinning an organization, building a digital and mirrored representation of its business processes is of paramount importance. Visualization techniques, including dashboards, are generally attributed to the effectiveness of a digital twin. Though dashboard engineering has attracted a great deal of attention in the last decade, it has been hampered by a lack of automation tools, leading to poorly designed dashboards that fail to grasp the analyst's analytical context. We develop a tool that semi-automatically creates a contextual dashboard that proves useful in the digital twin of a business process. Technically, our tool generates such a dashboard by iterating through all the pools and data objects visually depicted in a business process and generating the computer-interpretable source code that represents the appropriate visualization charts. To demonstrate the usefulness of our work, we walk through a real-life digital advertising project and illustrate how a dashboard generated by our tool could be displayed live using a dedicated visualization platform.



Dr. Lam-Son Lê

Lecturer in Software Engineering and
Programming Languages cum Academic
Coordinator for the Computer Science program

Speaker

BIO

Dr. Lam-Son Le (Member, IEEE) received his Ph.D. from the Swiss Federal Institute of Technology Lausanne (EPFL), Switzerland, in 2008. He spent four years as a Postdoctoral Researcher at the University of Wollongong, Australia. He currently holds dual roles at VGU: Academic Coordinator of the Bachelor's program in Computer Science and Lecturer in Software Engineering. His research interests include digital twins, decentralized computing, next-generation software architectures, and AI - driven business process management. Prior to joining VGU, he spent nine years at the Ho Chi Minh City University of Technology (HCMUT) - Vietnam National University, where he led a research group focused on data-driven enterprise software and processes. Lam-Son has successfully supervised three PhD students, served as program chair, demo track chair, publicity chair for various conferences in software engineering and enterprise computing. He has published around 70 papers in conference proceedings and journals.



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CONFERENCE SUMMARY

TBD